

Week 7 Cost-Benefit Memo - Revised

To: Dr. De Fretias, ED Board, and Martina Griffith, Clinical IT Lead

From: Vishal Baboolal

Subject: Phase 3 model choice for AI-assisted emergency triage

Date: 2026

1. Verdict

Advance the optimised LightGBM model to a silent Phase 3 pilot, retain logistic regression as the transparent reference, and keep both away from live clinical decisions.

2. Dataset and methods recap

The comparison used the same cleaned dataset and 80/20 stratified split from Week 6 (random_state=42): 55,121 encounters and 208 numeric pre-triage features. Post-triage outcomes, administrative identifiers, and fairness-sensitive demographic variables were excluded. Logistic regression and a bounded decision tree were retained; random forest, LightGBM, and XGBoost were screened. LightGBM was tuned on a training-only validation split, with the final test set reserved for evaluation.

3. Benchmark

Model	Accuracy	Macro F1	ESI 1 recall	Train (s)	Inference (ms)	Under 1 min?
Logistic regression	0.667	0.492	0.250	13.581	0.0103	YES
Random forest	0.497	0.398	0.500	7.301	0.0235	NO
LightGBM base	0.673	0.472	0.125	2.456	0.0325	NO
XGBoost base	0.614	0.377	0.188	3.566	0.0075	NO
Optimised LightGBM	0.664	0.518	0.250	3.908	0.0574	NO

The verdict asks whether the native model can support a meaningful patient-level explanation within one minute. Global feature importance alone does not satisfy that standard. Logistic regression passes; the ensemble models require a validated post-hoc explanation layer.

4. Arguments for the recommendation

- 1. **Best balance across ESI levels.** Macro F1 improved from 0.492 for logistic regression to 0.518 for optimised LightGBM, while overall accuracy remained competitive.
- 2. **Acceptable compute cost.** Measured training completed in 3.908 seconds and per-patient inference took 0.0574 milliseconds, which is operationally small for a silent pilot.
- 3. **Practical modelling capacity.** LightGBM can capture non-linear relationships and feature interactions while remaining easier to operate than a neural network.

5. Arguments against the recommendation

- 1. **Lower transparency.** Optimised LightGBM fails the native under-one-minute patient-level explanation test. A post-hoc tool and clinical review process would be required.
- 2. **Limited gain.** ESI Level 1 recall did not improve above logistic regression in this held-out set, so the added complexity has not yet shown better protection for the sickest group.
- 3. **Higher maintenance.** LightGBM adds a package dependency, more hyperparameters, model-version controls, explanation tooling, and monitoring requirements.

Interpretability decision record

Model family	Verdict	Basis
Logistic regression	YES	Coefficients and feature direction can be summarised in under one minute.
Single decision tree	YES	One prediction path can normally be traced in under one minute.
Random forest	NO	Hundreds of trees cannot be natively summarised for one patient in under one minute.
LightGBM / XGBoost	NO	Global importance is quick, but a patient-level explanation requires a post-hoc method.

6. Risks and unknowns

Critical-class uncertainty: only 16 ESI Level 1 cases were present in the held-out test set, so the recall estimate is unstable.

Infrastructure uncertainty: timing was measured on one runtime and must be repeated on Mercer infrastructure.

Clinical and governance gaps: fairness, calibration, drift, missing-data behaviour, alert burden, local Caribbean transferability, and explanation quality remain untested.

Interpretability gap: global feature importance does not explain why one patient received one specific ESI prediction.

7. Recommendation

Proceed only with a silent pilot beside logistic regression. Compare critical-class errors, subgroup performance, calibration, latency, and explanation usability on local data. Do not expose predictions to triage staff or use them for prioritisation until the evidence is stronger and the explanation approach is clinically approved.

Audit links: See docs/model-selection.md for all six comparison axes and docs/decisions/2026-week-7-model-choice.md for the decision journal.

This memo is a research handover document, not a clinical deployment approval.